String:-

*The string is a **collection of alphabets, words, or other characters**. It is one of the primitive data structures and are the building blocks for data manipulation.*

Assign String to a Variable

Single Word

```
a = "Hello"
a = 'Hello'
a = "10"
a = '1'


print(a)
```

Multi-line String(Example 01)

```
a = """We invite you to attend the Best IT Course at our institute. We are happy to guide you step by step regarding this job-oriented course and the job placement benefits after completing the course."""
print(a)
```

Multi-line String(Example 02)

```
a = ''' We invite you to attend the Best IT Course at our institute. We are happy to guide you step by step regarding this job-oriented course and the job placement benefits after completing the course.'''
print(a)
```

Strings are Arrays

In Python, a string is a sequence of characters enclosed within either single quotes (') or double quotes (").

Slicing in string

```
b = "Hello, World!"
print(b[2:])
print(b[2:5])
print(b[:5])
print(b[-6:])
print(b[-5:-2])
```

```
llo, World!
llo
Hello
World!
orl
```

Looping through string

```
for x in "banana":
  print(x)
```

```
b
a
n
a
n
a
```

Check the length of the string

```
a = "Hello, World!"
print(len(a))
```

```
13
```

Check specific text(I.e. free) in the string

```
txt = "The best things in life are free!"
print("free" in txt)
```

```
True
```

```
txt = "The best things in life are free!"
if "free" in txt:
  print("Yes, 'free' is present.")
```

```
Yes, 'free' is present.
```

Check if "expensive" is NOT present in the following text:

```
txt = "The best things in life are free!"
print("expensive" not in txt)
```

```
True
```

```
txt = "The best things in life are free!"
if "expensive" not in txt:
  print("Yes, 'expensive' is NOT present.")
```

```
Yes, 'expensive' is NOT present.
```

Escape Character

Operation	Code	Example	Result	Remarks
Single Quote	\'	txt = 'It\'s alright.'	It's alright.	
Backslash	\\	txt = "Hello\\World!"	"Hello\World!"	This will insert one \ (backslash).
New Line	\n	txt = "Hello\nWorld!"	Hello World!	
Carriage Return	\r	txt = "Hello\rWorld!"	Hello World!	
Tab	\t	txt = "Hello\tWorld!"	Hello World!	
Backspace	\b	txt = "Hello \bWorld!"	HelloWorld!	#This example erases one character (backspace):
Octal value	\ooo	txt = "\110\145\154\154\157"	Hello	#A backslash followed by three integers will result in an octal value:
Hex value	\xhh	txt = "\x48\x65\x6c\x6c\x6f"	Hello	#A backslash followed by an 'x' and a hex number represents a hex value: